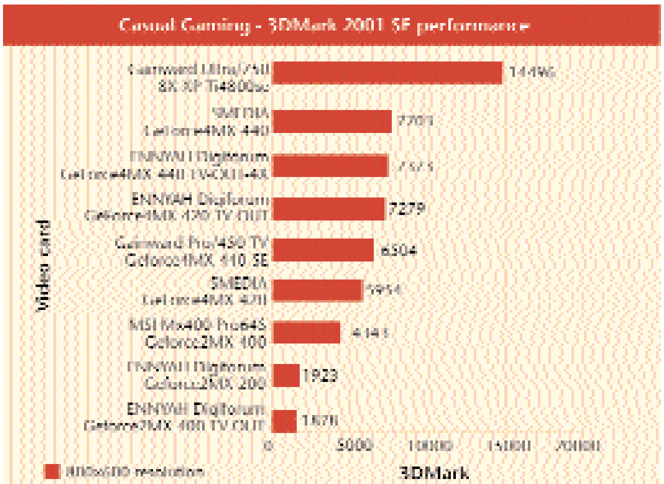

and greater than the 6,500 mark under 3DMark2001 SE. They do similarly well across the board—their performance under *Serious Sam* hardly falls under the 80 fps mark, while even the stressful *UT 2003* never manages to bring them to a level less than the 50 fps level. Consider the GeForce4MX 440-4X and the 420 variants: at Rs 4,500 and Rs 4,750, respectively you can see why it is wiser to invest in one such chipset rather than one based on the GeForce2MX chipset. Cheaper variants of the chipset are on offer by SMEDIA cards—the GeForce4MX 420 and the GeForce4MX 440 by SMEDIA ask for Rs 3,200 and Rs 3,900, respectively. Although the SMEDIA solutions do not perform up to the par set by the ENNYAH cards, the performance married with the prices on offer allow us to consider them as a better buy. A recommendation that is reflected by the STTI score that these cards achieve: 81.80 STTI for the SMEDIA GeForce4MX 420 and 88.68 STTI for the GeForce4MX 440 variant. Judged purely on performance, the Gainward Ultra/750 8X XP Ti4800SE comes across as the leader. Colour it with a price of Rs 18,895, however, and you can see why the SMEDIA cards are the overall winners in this category.

Our recomended solution: SMEDIA GeForce4MX 420

High-end gaming

Although the golden rule of greater than 30 fps still holds for the high-end gaming segment, the landscape changes drastically due to several factors. First amongst them is the resolution—when you find yourself shelling out money in excess of Rs 10,000, your expectations should similarly rise. Thus, frame rates of around 60 fps at resolutions of up to 1280x1024 are considered noteworthy. Similarly, a score around the 15,000 3DMark should be considered ideal for such a card. Once resolution is duly handled, the 3D quality offered by a card is evaluated. For this, we concentrated on the visuals that a card dished out under the later and the greater engine of *UT 2003*, giving it four times as much weightage as we gave *Quake III Arena*. To do this, we enabled Full Screen Anti-Aliasing (FSAA) and Anisotropic filtering for all the cards. A panel of three then judged these and awarded points. We also sampled FSAA scores for these cards, in order to gauge the performance hit that each one took while enabling the said feature.

Finally, benchmarks of *Serious Sam*, 3DMark2001 SE and *UT 2003* were given the maximum weightage for performance under this application area. Feature-wise, we judged each card on dual-monitor support, I/O ports (such as S Video-RCA), memory bandwidth, number of pixel/vertex pipelines, API compatibility, etc.

The best video cards for High-End Gaming: A performance STTI of 66 along with an ideal price of Rs 10,000 came together to define the ideal buy under this particular category. To get a range of suitable cards, we formed a band centred at the ideal card with price ranging from a low of Rs 8,000 to a high of Rs 12,000. Thus, all cards with a score greater than 66 STTI and falling within the above price band were deemed suitable to buy. Even though the performance scores within this area were skewed by a relatively low price factor of 20 per cent, it was enough to cause some cards to lose to cheaper brethren. A card such as the Matrox Parhelia was thus immediately out of the running—this particular variant cost a staggering Rs 30,000 even as it failed to score the 66 STTI needed. Similarly upstaged were the Gainward Ultra/750 8X XP Ti4800SE and the Hercules 3D PROPHET R9700. While the Gainward scored 79.22 STTI, its price of Rs 18,895 humbled it next to the Gainward Ultra/650 TV GeForce4 Ti4200—the latter scored significantly less at 78.79 STTI but also came significantly cheaper at Rs 11,995. An identical fate awaited the 3D PROPHET—the R9700-based card scored 75.6 STTI, but its high price of Rs 24,895 saw it lose its glory to another R9700 card—the SMEDIA RADEON 9700 128 MB. The latter scored 82.11, thanks to its cheaper tag of Rs 16,000.

That price is a significant leveller can be further ascertained by looking at the best performers in this application area. The Hercules 3D Prophet R9700 PRO scored 74.04 STTI, while the SMEDIA RADEON 9700 128MB scored 82.11 STTI. Factor in price and the picture becomes clear: between the Hercules at Rs 33,500 and the XSERVE at Rs 16,000, the choice becomes obvious—the latter becomes a recommended solution for the task at hand.

When it came to pure numbers, the ATI cards proved to be a superior buy. From nVidia's stable, the GeForce4MX chipset made a repeat appearance. As pointed out earlier, the low cost and good enough feature set of these chipsets make them an ideal buy—although not one with a great lifespan—for the non-discerning gamer. Solutions from ASUS, COMPRO, Gainward and MSI carried variants of this very chipset.

While all of them managed to cross the 50 fps mark in the *Serious Sam* test, nothing beat the Gainward Pro/600-8X XP GeForce4MX 440 as it touched almost 60 fps at 1280x1024, a great showing indeed considering that it came in as the cheapest buy at Rs 8,300. It also did equally well in the DirectX stresser, *UT 2003*, posting a very playable 42.85 fps at 1280x1024. What then took away its best buy status? The XSERVE RADEON 9700. Its amazing posting of 82.11 STTI at a price of Rs 16,000, tilted the scales in its